

# SISTEM DIGITAL (TIF-1106)

Pertemuan 12: Memori (RAM, ROM) & Programmable Logic (PLA, PAL, FPGA)

# Capaian Pembelajaran Pertemuan 12

## Sub-CPMK 12

Mahasiswa mampu memahami arsitektur internal memori (RAM SRAM/DRAM, ROM), mendekode bus alamat (\*address decoding\*), serta menganalisis perangkat logika terprogram (\*Programmable Logic Devices\*: PLA, PAL, CPLD, dan FPGA).

# Perbandingan Perangkat Logika Terprogram

| Tipe Perangkat                       | Matriks AND                        | Matriks OR                         |
|--------------------------------------|------------------------------------|------------------------------------|
| PROM                                 | Tetap (Fixed Decoder)              | Terprogram ( <i>Programmable</i> ) |
| PLA (Programmable Logic Array)       | Terprogram ( <i>Programmable</i> ) | Terprogram ( <i>Programmable</i> ) |
| PAL (Programmable Array Logic)       | Terprogram ( <i>Programmable</i> ) | Tetap (Fixed)                      |
| FPGA (Field Programmable Gate Array) | Berbasis *Look-Up Table* (LUT)     | Blok Logika Terkonfigurasi (CLB)   |